Taek Lim

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SUMMARY

Software engineer with about 5 years across the data lifecycle: collection, streaming ingestion, storage, and visualization, on a cloud recommendation platform on AWS (600M+ logs monthly) and a multi-channel ad-analytics SaaS. MS Computer Science at UTRGV (05/2027), researching generative AI and assisting a Deep Learning course. Seeking software and data engineering roles.

PROFESSIONAL EXPERIENCE

Senior Software Engineer, ADRIEL **Seoul, South Korea, 11/2021–05/2024**

- Cut most dashboard loads from 3s to under 1s via a **React/Next.js** migration, **RxJS** event control, and a server-aggregation API redesign.
- Traced slow dashboards and wrong totals to browser-side aggregation and proposed moving aggregation and caching to the server.
- Reduced client report generation time from 2 hours to 1 hour by developing a PDF export feature with layout conversion algorithms.

Software Developer, DIV **Seoul, South Korea, 05/2021–10/2021**

- Integrated hotel-booking into a rent-a-car platform and migrated flight-booking to a new broker, lifting revenue by 3%.

Software Engineer, BABELTOP **Seoul, South Korea, 07/2019–11/2019**

- Built translator-approval, login, sign-up, and resume features for a translation SaaS using **React**, **Redux**, and **Redux-Saga**.

Software Engineer, RECOBELL **Seoul, South Korea, 02/2018–07/2019**

- Operated a cloud-based recommendation system using **Java**, **Spring**, **MySQL**, and **AWS**, processing 600M+ user logs monthly.
- Built keyword-based book recommendation APIs for Korea's No. 1 bookstore ($500M annual revenue), handling 1M+ daily requests.
- Maintained and extended data pipelines, a **SQL**-based recommendation engine, and APIs using **AWS Kinesis**, **S3**, and **Java**.

RESEARCH AND TEACHING

Graduate Research and Teaching Assistant, UTRGV **Edinburg, TX, 08/2026–Present**

- Generating nocturnal bird flight calls from 30–130 samples using open-source audio models to support rare-species research.
- As Teaching Assistant, built a **PyTorch** DDPM on SLURM, improving public FID from 30.26 to 6.10 under a 10M-parameter cap.
- Built a **PyTorch** ResNet ensemble reaching 93.2% private accuracy on 100 classes under a 5M-parameter cap, across 158 training runs.

PROJECTS

Weather and BTC Prediction-Market Data Pipeline: TypeScript, Kafka, ClickHouse, k3s **04/2026–Present**

- Collected Kalshi, Polymarket, and 60-city weather data into ClickHouse on a Mac mini, with 39 Kafka topics and k3s on Oracle Cloud.
- Deduplicated rows with idempotency keys and ClickHouse ReplacingMergeTree under one Protobuf schema.

Personal Blog and Portfolio: Next.js, Express, Postgres, GCP, OCI, Cloudflare **10/2025–Present**

- Built a Python publisher from an NFS-mounted home NAS to Postgres and Oracle Object Storage, with Claude-generated tags.
- Restored a suspended site and cut Supabase egress from 40+ GB a week to 2.47 GB in October using GCP and Oracle storage at $0.
- Traced 16 traffic causes and added Cloudflare caching and rate limiting: a 640-request test on cached URLs reached the VM 0 times.

MapReduce Infrastructure Implementation: C++, gRPC, protobuf **11/2024**

- Implemented a simplified distributed MapReduce framework with asynchronous gRPC, based on the Dean and Ghemawat paper.

SKILLS

- Languages: Java, Python, C/C++, TypeScript/JavaScript, SQL
- Data: Kafka, AWS Kinesis, PostgreSQL/TimescaleDB, Supabase, MySQL, ClickHouse, Redis, Grafana
- ML: PyTorch, TensorFlow, pandas, NumPy, SLURM
- Backend: Spring Boot/Security, JPA, MyBatis, FastAPI, Express, Node, Electron, REST/GraphQL, gRPC/Protobuf, JUnit
- Cloud: AWS (EC2/S3/RDS/EBS), Azure, GCP, Oracle Cloud, Cloudflare, Vercel, Clerk
- DevOps: Docker, Kubernetes (k3s), Kustomize, GitHub Actions, k6, Tailscale, Git, Gradle, Maven, Webpack, Jest

EDUCATION

The University of Texas Rio Grande Valley **Edinburg, TX**
MS Computer Science, Graduate Select Scholarship *05/2027*

University of Central Missouri **Warrensburg, MO**
MS Computer Science, Cybersecurity and Information Assurance, Graduate Student Achievement Award *01/2025–05/2026*
Graduate Assistant *01/2026–05/2026*

University of California, Los Angeles **Los Angeles, CA**
BS Mathematics-Economics *03/2013*

CERTIFICATIONS AND LICENSES

- AWS Certified AI Practitioner 08/2026
- SQL Developer Certificate, Korea Data Agency 07/2023
- CS Engineer Information Processing License, HRDK 11/2018